DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Subject Name: **WEB TECHNOLOGY** Subject Code: **CS T63**

UNIT-IV

Multimedia and Web Application: Multimedia in web design, Audio and video speech synthesis and recognition - Electronic Commerce – E-Business Model – E-Marketing – Online Payments and Security – N-tier Architecture. Search and Design: Working of search engines -optimization-Search interface.

2 MARKS

1. Define Multimedia?

Multimedia includes a combination of text, audio, still images, animation, video, and interactivity content forms.

2. List the applications of Multimedia?

- Instruction
- Video conferencing
- Digital Transmission
- Groupware
- Games
- Digital video editing and production systems
- Scientific analysis, research and presentations
- Business applications, advertising and sales
- Adaptive technologies for the disabled
- Virtual reality

3. What are the common audio file format?

- Audio and video can be used in Web pages in a variety of ways. Audio and video files can be embedded in a Web page or placed on a Web server such that they can be downloaded "on-demand." A variety of audio and video file formats are available for different uses.
  - MP3 (MPEG Layer 3)
  - MIDI (Musical Instrument Digital Interface)
  - WAV (Windows Waveform) and
  - AIFF (Audio Interchange File Format - Macintosh only)

4. What are the common video file format?

- Quick – Time
- RealPlayer
- Windows media file (.asf)
- Windows video file (.avi)
- Movie file (.mpeg)
- Windows media video file (.wmv)

5. What is SVG?

Scalable Vector Graphics (**SVG**) is an XML-based vector image format for two-dimensional graphics with support for interactivity and animation. SVG images and their behaviors are defined in XML text files. This means that they can be searched, indexed, scripted, and compressed. As XML files, SVG images can be created and edited with any text editor, but are more often created with drawing software.

6. List the various properties of bgsound elements?

The bgsound element has four key properties –

- ✓ src,
- ✓ loop,
- ✓ balance and
- ✓ volume

7. List the different steps to be followed while adding Multimedia file to the web? (APR 2016)

- The <AUDIO> tag is used to include an audio file in a Web page.
- The <VIDEO> tag is used to display a video file on a Web page.

8. What is the use of image element? What are its properties?

The img element enables both images and videos to be included in a Web page. The src property indicates that the source is an image. The dynsrc (i.e., dynamic source) property indicates that the source is a video clip. Property start indicates when the video should start playing (specify file open to play when the clip is loaded or mouseover to play when the user first positions the mouse over the video).

9. What is the function of embed element?

The embed element embeds a media clip in a webpage. A GUI can be displayed to give the user control over the media clip and it enables the user to play, pause & stop the media clip to specify the volume and to move forward and backward quickly through the clip.

10. What is the purpose of e-business?

An e-business can offer personalization, effective customer service & streamlined supply chain management.

11. What is e-commerce?

Electronic commerce commonly known as e-commerce, is the buying and selling of products or service over electronic systems such as the internet and other computer.

E-Commerce is a methodology of modern business which addresses the need of business organizations, vendors and customers to reduce cost and improve the quality of goods and services while increasing the speed of delivery.

12. What are the various e-commerce technology?

- Electronic Data Exchange (EDI)
- Electronic Mail (e-mail) or Internet Marketing
- Electronic Bulletin Boards
- Electronic Fund Transfer (EFT)
- Supply chain management
- Online transaction processing
- Inventory Management Systems and
- Automated data collection systems

13. What are the features of e-commerce? **(APR 2017)**

- Non-Cash Payment
- 24x7 Service availability
- Advertising / Marketing
- Improved Sales
- Support
- Inventory Management
- Communication improvement

14. List out the basic security requirements for E-Business? **(APR 2016)**

- Integrity: prevention against unauthorized data modification
- Nonrepudiation: prevention against any one party from reneging on an agreement after the fact
- Authenticity: authentication of data source
- Confidentiality: protection against unauthorized data disclosure
- Privacy: provision of data control and disclosure
- Availability: prevention against data delays or removal

15. List the e-commerce threats?

- Loss of Privacy/confidentiality, data misuse/abuse
- Cracking, eavesdropping, spoofing, rootkits
- Viruses, Trojans, worms, hostile ActiveX and Java
- System unavailability, denial of service, natural disasters, power interruptions

16. What is meant by EDI?

Electronic Data Interchange (EDI) standardizes business forms, such as purchase orders and invoices, so that companies can share information electronically with customers, vendors and business partners.

17. What are the advantages and disadvantages of e-commerce? Advantages

1. No Geographic limitations
2. Savings
3. 24*7 business
4. Better Productivity
5. Quick Comparison
6. Economy Benefit

Disadvantages

1. Security/ Privacy
2. Quality
3. Reliability
4. Products People
5. Customer Service and Relation Problem

18. What are the different types of e-Business model?

- Business-to-Consumer (B2C)
- Business-to-Business (B2B)
- Consumer - to - Consumer (C2C)
- Consumer - to - Business (C2B)
- Business - to - Government (B2G)
- Government - to - Business (G2B)
- Government - to - Citizen (G2C)

19. What are the Major Business-to-Consumer (B2C) Business Models?

- Portal
- E-tailer
- Content Provider
- Transaction Broker
- Market Creator

20. What are the Major Business-to-Business (B2B) Business Models?

- B2B Hub
- E-distributor
- B2B Service Provider
- Matchmaker
- Infomediary

21. What are the Components of e-Business Models?

- Price Competition
- E-Commerce Business Models
- Value Position
- Revenue Model
- Market Opportunity
- Competitive Environment
- Market Strategy
- Organizational Development
- Management Team

22. Define e-Marketing? Mention any one use of it. **(NOV 2016)**

- e-Marketing is any marketing done online via websites or other online tools and resources.
- e-Marketing can include paid services while other methods are virtually free.
- e-M arketing methods are at your disposal, including: direct email, SMS/text messaging, blogs, webpages, banners, videos, images, ads, social media, search engines etc.

The application of e-marketing

- Reduced Time & Effort
- Personalize Messages
- More Frequent Communications
- Information Spreading
- Reduce Overhead Costs

23. List the types of e-marketing?

- Transactional emails
- Direct emails
- Mobile email marketing

24. What are the different 7 Cs of E-Marketing?

- Contract
- Content
- Construction
- Community
- Concentration
- Convergence
- Commerce

25. List the various e-Marketing Methods?

- Search engine marketing
- Display Advertising
- E-mail marketing
- Interactive marketing
- Blog marketing
- Viral marketing

26. Mention the advantages and disadvantages of e-marketing? (NOV 2017)

Advantages

- Availability of Information
- Saves money
- Expansion
- Low Cost
- Efficiency of Advertising

Disadvantages

- Technology
- Low connection speed
- Complication
- Intangibility

27. What is meant by SSL?

The Secure Sockets Layer (SSL) protocol is commonly used to secure communication on the Internet and the Web. SSL uses public-key technology and digital certificates to authenticate the server in a transaction and to protect private information as it passes from one party to another over the Internet.

28. What is e-Payment?

- Electronic payment at any time, through the Internet directly to the transfer, settlement and form e-business environment. To ensure the safety and secrecy of the parties to a transaction, which requires a complete electronic trading systems.
- The system that permits online payment between parties using an electronic surrogate of a financial tender. The intent is to act as an alternative form of payment to the physical cash, cheque or other financial tender.

29. List the various e-payment transactions?

- Credit Card
- Debit Card
- Smart Card
- E-Money
- E-Wallets
- Electronic Fund Transfer (EFT)

30. How to securing e-payments?

- Identification and authenticate
- Authorization
- Integrity and confidentiality
- Accountability
- Policies for sharing risks and liabilities

31. What are the essential requirements for e-payment/transactions?

- Confidential
- Integrity
- Availability
- Authenticity
- Non-Repudiabiity
- Encryption
- Auditability

32. What is web crawling?

Are search engines that use software programs called that are called “Spiders”, “crawlers”, “Robots”, “Bots”. These programs can access the web pages to categorize and analyze them and then add them in the search engine database, where any user can find then when searching. The crawler based search engines are constantly updated with new web page that would be available in their database.

Examples:

Google

Yahoo

Ask.com

33. What are the designing a search interface? (NOV 2016)

- Search form must fit the type of data being searched.
- HTML and Scripting languages are used.
- Primary element – Search query field.
- Second aspect – Button to execute the search. (Form buttons or Custom buttons).

34. What is Search engine? (APR 2017)

- An internet-based tool that searches an index of documents for a particular term, phrase or text specified by the user. Commonly used to refer to large web-based search engines that search through billions of pages on the internet.
- A search tool that allows a user to enter queries. The program responds with a list of matches from its database. A relevancy score for each match and click able URL is usually returned.

35. Write the examples of search engines?

- Conventional
- Text-based
- Multimedia
- Question answering systems
- Clustering systems
- Research systems

36. Mention the common characteristics of search engines?

- Spider, Indexer, Database, Algorithm
- Find matching documents and display them according to relevance
- Frequent updates to documents searched and ranking algorithm
- Strive to produce “better”, more relevant results than competitors

37. List the various search engines?

- Crawler based search engines
- Directories
- Specialty search engines
- Hybrid search engines
- Meta search engines

38. Define SEO?

Search engine optimization (SEO) is the process of improving the visibility of a website or a web page in search engines via the "natural," or "algorithmic", search results.

39. Mention the various SEO techniques?

- Domain name strategies
- Linking strategies
- Keywords
- Title tags
- Meta description tags
- Alt tags

40. What is Multi-tier Application?

A Web server typically is part of a multi-tier application – sometimes referred to as an n-tier application. A multi-tier application divides functionality into separate tiers. The three-tier application contains an information tier, a middle tier and a client tier.

41. What is ActiveX Control? **(APR 2016)**

- ActiveX is a software framework created by Microsoft that adapts its earlier Component Object Model (COM) and Object Linking and Embedding (OLE) technologies for content downloaded from a network, particularly in the context of the World Wide Web.
- ActiveX is a software framework for defining reusable software components in a programming language – independent way (ie. not tied to a particular programming language).
- ActiveX controls to build their feature-set and also encapsulate their own functionality as ActiveX controls which can then be embedded into other applications. Internet Explorer also allows the embedding of ActiveX controls in web pages.

42. Write in short about Streaming in web applications. **(NOV 2017)**

Web Application Streaming is a radical new approach to sending web sites and applications to standard web browsers which provides for dramatically faster page load times.

11 MARKS

1. Write a short note on Multimedia in web design? **(5 MARKS) (APR 2017)**

Multimedia includes a combination of text, audio, still images, animation, video, and interactivity content forms.

An interactive computer-mediated presentation that includes at least two of the following elements:

- Text
- Sound
- Still graphic images
- Motion graphics
- Animation

Multimedia computers serve two primary functions:

- Development tools for the creation of multimedia productions
  - Macromedia – Flash, Dreamweaver, Director etc
  - Real Systems Producer Plus
  - CoolEdit
- Delivery devices for multimedia presentations
  - Browser plug-ins or player applications

- Multimedia files can be quite large. Some multimedia technologies require that the complete multimedia file be downloaded to the client before the audio or video begins playing. With streaming technologies, audio and video can begin playing while the files are downloading, to reduce delays. Streaming technologies are becoming increasingly popular.
- Creating audio and video to incorporate into Web pages often requires complex and powerful software such as Adobe™ After Effects or Macromedia™ Director. Rather than discuss how to create media clips, this chapter focuses on using existing audio and video clips to enhance Web pages. The chapter also includes an extensive set of Internet and World Wide Web resources. Some of these Web sites display examples of interesting multimedia enhancements; others provide instructional information for developers planning to enhance their own sites with multimedia.

Applications of Multimedia

- Instruction
- Video conferencing
- Digital Transmission
- Groupware
- Games
- Digital video editing and production systems

- Scientific analysis, research and presentations
- Business applications, advertising and sales
- Adaptive technologies for the disabled
- Virtual reality

2. Explain briefly about audio and video speech synthesis and recognition? (11 MARKS) (NOV 2016)

Audio and Video:

- Audio and video can be used in Web pages in a variety of ways. Audio and video files can be embedded in a Web page or placed on a Web server such that they can be downloaded "on-demand." A variety of audio and video file formats are available for different uses.

File formats

Video

- ✓ MPEG (Moving Pictures Experts Group)
- ✓ QuickTime
- ✓ RealPlayer
- ✓ AVI (Video for Windows)
- ✓ MJPEG (Motion Joint Photographic Experts Group)

Audio

- ✓ MP3 (MPEG Layer 3)
- ✓ MIDI (Musical Instrument Digital Interface)
- ✓ WAV (Windows Waveform)
- ✓ AIFF (Audio Interchange File Format) Macintosh only

Encoding and compression determine a file's format. An **encoding algorithm** or CODEC compresses media files by taking the raw audio or video and transforming it into a format that Web pages can read. Different encoding levels and formats produce file sizes that are ideal for different applications.

Some CODECs are available to the public in the form of **encoding applications**. Most encoding applications compress audio and video files. Some serve as **format converters**, converting one file format into another.

Adding Background Sounds with the bgsound Element:

- Web sites provide to create background audio.
- Various ways exist to add sound to a Web page, the simplest is the bgsound element.

Syntax:

```
<bgsound src = "url" loop = "-1 | 0" balance = "0 | -10000 | 10000" volume = "-10000 | 0" >
</bgsound>
```

Properties of bgsound element:

1. The **src property** specifies the URL of the audio clip to play. Internet Explorer supports a wide variety of audio formats.
2. The **loop property** specifies the number of times the audio clip will play. The value -1 (the default) specifies that the audio clip should loop until users browse a different Web page or click the browser's Stop button. A positive integer indicates the exact number of times the audio clip should loop. Negative values (except -1) and zero values for this property cause the audio clip to play once.
3. The **balance property** specifies the balance between the left and right speakers. The value for this property is between -10000 (sound only from the left speaker) and 10000 (sound only from the right speaker). The default value, 0, indicates that the sound should be balanced between the two speakers.
4. The **volume property** determines the volume of the audio clip. The value for this property is between 10000 (minimum volume) and 0 (maximum volume). The default value is 0.

The volume specified with bgsound property volume is relative to the current volume setting on the client computer. If the client computer has sound turned off, the volume property has no effect.

Example:

```
<html>
        <head>
                <title> Bgsound Elements </title>
                <bgsound src="veeram.mp3" loop="-1"
        balance="0"> </head>
</html>
```

Adding Video with the img Element's dynsrc Property:

Users can tremendously enhance the multimedia presentations by incorporating a variety of video formats into their Web pages.

Syntax:

```
<img dynsrc = "url" start = "fileopen | mouseover" width = "180" height = "135" loop = "-1"
 alt = "text" align="right | middle | left | top | bottom border = "2" >
```

Properties of img element:

1. The img element incorporates both images and videos in a Web page. The src property indicates that the source is an image.
2. The dynsrc (i.e., dynamic source) property indicates that the source is a video clip. The dynsrc property may have other properties such as loop, which is similar to the bgsound loop property.

3. The start property specifies when video should start playing
    - Fileopen - Video should play as soon as it loads
    - Mouseover - Video should play when user position mouse over video

Example:

```
<html>
        <head>
                <title> An Embedded Video Using the dynsrc Property </title>
                < bgsound src ="Theri.mp3" loop = "-1"> </bgsound>
        </head>

<body>
        <h1>Theri Theme Video </h1>
        <table>
                <tr>
                        <td> <img dynsrc = "Theri.mp4" start = "mouseover" width = "180" height = "135" loop
                        = "-1" alt = "Car driving in circles" > </td>
                        <td> This page will play the audio clip and video in a loop. <br >
                                The video will not begin playing until you move the mouse over the video. <br
                                > Press <strong> Stop </strong> to stop playing the sound and the video.
                        </td>
                </tr>
        </table>
</body>
</html>
```

Output:

Adding Audio or Video with the embed Element:

- The embed element, which embeds a media clip (audio or video) into a Web page. The embed element displays a graphical user interface that gives users direct control over the media clip. When the browser encounters a media clip in an embed element, the browser plays the clip with the player registered to handle that media type on the client computer.
- For example, if the media clip is a wave file (i.e., a Windows Wave file), Internet Explorer typically uses the Windows Media Player ActiveX control to play the clip.
- The Windows Media Player has a GUI that enables users to play, pause and stop the media clip. Users can also control the volume of audio and move forward and backward through the clip using the GUI.
- The embed element is supported by both Microsoft Internet Explorer and Netscape navigator.

Syntax:

embed src = "filename" autostart = "true | false" hidden = "true | false" loop = "true | false"

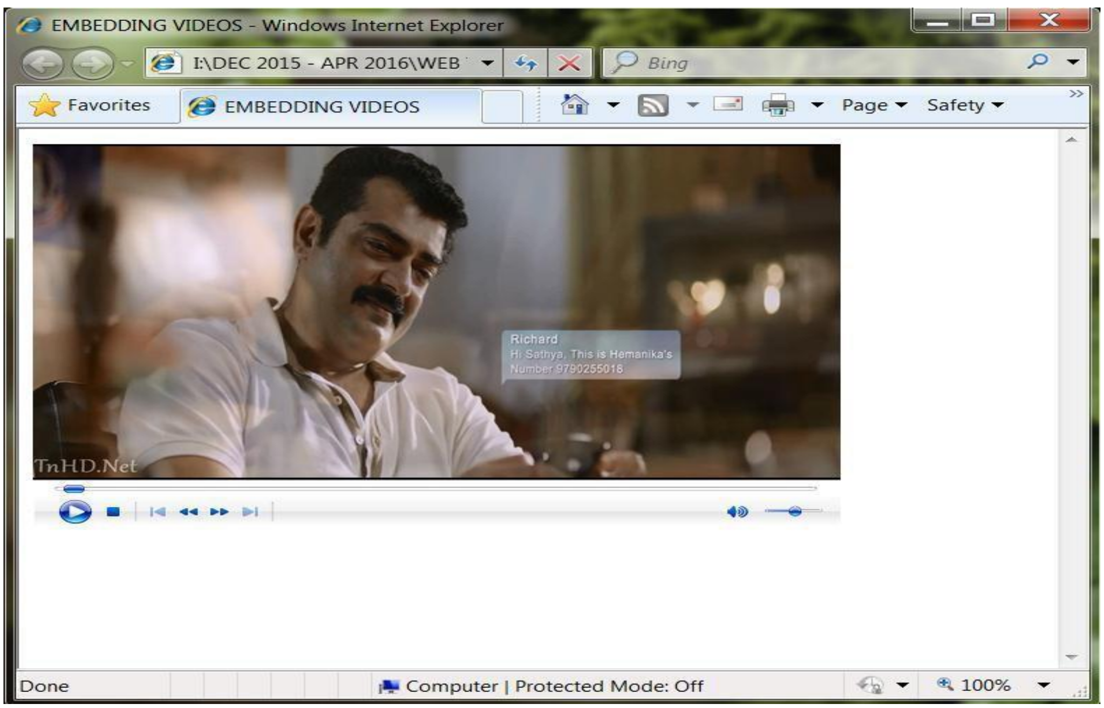
height = "50%" width = "50%" align="right | middle | left | top | bottom > </embed>

- The hidden property
  - ✓ Prevent GUI from being displayed
  - ✓ GUI displayed by default
- The LOOP property- Loop clip forever

Example:

```
<html>
        <head>
                <title> Embedding Videos </title>
        </head>
        <body>
                <embed src = "Mazhai.mp4" autostart = "true" loop = "true" height = "75%" width = "80%"
                hidden = "false">
                </embed>
        </body>
</html>
```

Output:



Using the Windows Media Player ActiveX Control

ActiveX controls enhance the functionality of Web pages with interactivity. To embed the Windows Media Player ActiveX control in a Web page, so that we can access a wide range of media formats supported by the Windows Media Player. The Windows Media Player and other ActiveX controls are embedded into Web pages with the object element.

Syntax:

```
<OBJECT ID = "VideoPlayer" WIDTH = " 200 " HEIGHT = " 225 " CLASSID = " CLSID:22d6f312-b0f6-11d0-
  94ab-0080c74c7e95" >
      <PARAM NAME = "FileName" VALUE = "Theri.mp4">
      <PARAM NAME = "AutoStart" VALUE = "true">
      <PARAM NAME = "ShowControls" VALUE = "false">
      <PARAM NAME = "Loop" VALUE = "true">
</OBJECT>
```

Property:

- Property id
  - ✓ Specifies scripting name of element
- width and height properties
  - ✓ Specify width and height in pixels
- property classid
  - ✓ Specifies ActiveX control ID

Element Param

- FileName
  - ✓ Specifies file containing media clip
- AutoStart
  - ✓ Boolean for whether clip should start playing when it loads
- ShowControls
  - ✓ Boolean for whether controls should be displayed
- Loop
  - ✓ Boolean for whether clip should loop infinitely
- Methods:
  - ✓ Play
  - ✓ Pause

Example:

```
<HTML>
	<HEAD>
		<TITLE> Embedded Media Player Objects </TITLE>
		<SCRIPT LANGUAGE = "JavaScript">
			var videoPlaying = true;
			function toggleVideo( b )
			{
				videoPlaying =! videoPlaying;
				b.value = videoPlaying ? "Pause Video1" : "Play Video2";
				videoPlaying ? VideoPlayer.Play() : VideoPlayer.Pause();
			}
		</SCRIPT>
	</HEAD>
		<BODY>
		<H1 ALIGN="CENTER"> Audio and video through embedded Media Player objects </H1>
		<HR COLOR="RED" SIZE="5">
		<TABLE ALIGN="CENTER">
			<TR>
				<TD VALIGN = "top" ALIGN = "center">
	<OBJECT NAME = "VideoPlayer" WIDTH ="400" HEIGHT ="425" CLASSID ="CLSID:22d6f312-
	b0f6-11d0-94ab-0080c74c7e95">
				<PARAM NAME = "FileName" VALUE = "Theri.mp4">
				<PARAM NAME = "AutoStart" VALUE = "true">
				<PARAM NAME = "ShowControls" VALUE = "false">
				<PARAM NAME = "Loop" VALUE = "true">
	</OBJECT>
				</TD>
				<TD VALIGN = "bottom" ALIGN = "center">
					<P>Use the controls below to control the audio clip.</P>
		<OBJECT ID = "AudioPlayer" CLASSID = "CLSID:22d6f312-b0f6-11d0-94ab-0080c74c7e95">
					<PARAM NAME = "FileName" VALUE = "Theri.mp3">
					<PARAM NAME = "AutoStart" VALUE = "true">
					<PARAM NAME = "Loop" VALUE = "true">
		</OBJECT>
				</TD>
			</TR>
```

```
                <TR>
                        <TD VALIGN = "top" ALIGN = "center">
        <INPUT NAME = "video" TYPE = "button" VALUE = "Pause Video" ONCLICK =
                        "toggleVideo(this)"> </TD>
                </TR>
            </TABLE>
            <HR COLOR="RED" SIZE="5">
        </BODY>
</HTML>
```

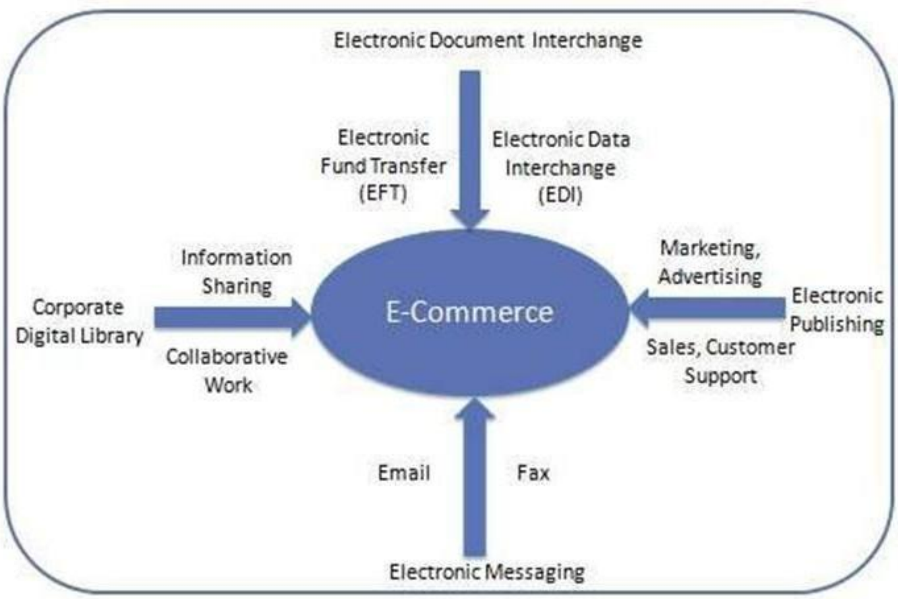
Output:

3. Explain briefly about electronic commerce? **(11 MARKS) (NOV 2017)**

- Electronic commerce commonly known as e-commerce, is the buying and selling of products or service over electronic systems such as the internet and other computer.
- E-Commerce or Electronics Commerce is a methodology of modern business which addresses the need of business organizations, vendors and customers to reduce cost and improve the quality of goods and services while increasing the speed of delivery.

E-commerce refers to paperless exchange of business information using following ways.

- Electronic Data Exchange (EDI)
- Electronic Mail (e-mail) or Internet Marketing
- Electronic Bulletin Boards
- Electronic Fund Transfer (EFT)
- Supply chain management
- Online transaction processing
- Inventory Management Systems and
- Automated data collection systems



Features

E-Commerce provides following features

1. **Non-Cash Payment** – E-Commerce enables use of credit cards, debit cards, smart cards, electronic fund transfer via bank's website and other modes of electronics payment.
2. **24x7 Service availability** – E-commerce automates business of enterprises and services provided by them to customers are available anytime, anywhere. Here 24x7 refers to 24 hours of each seven days of a week.
3. **Advertising / Marketing** – E-commerce increases the reach of advertising of products and services of businesses. It helps in better marketing management of products / services.
4. **Improved Sales** – Using E-Commerce, orders for the products can be generated anytime, anywhere without any human intervention. By this way, dependencies to buy a product reduce at large and sales increases.
5. **Support** – E-Commerce provides various ways to provide pre sales and post sales assistance to provide better services to customers.
6. **Inventory Management** – Using E-Commerce, inventory management of products becomes automated. Reports get generated instantly when required. Product inventory management becomes very efficient and easy to maintain.
7. **Communication improvement** – E-Commerce provides ways for faster, efficient, reliable communication with customers and partners.

Advantages

E-Commerce advantages can be broadly classified in three major categories:

1. Advantages to Organizations
2. Advantages to Customers
3. Advantages to Society

Advantages to Organizations

- Using E-Commerce, organization can expand their market to national and international markets with minimum capital investment. An organization can easily locate more customers, best suppliers and suitable business partners across the globe.
- E-Commerce helps organization to reduce the cost to create process, distribute, retrieve and manage the paper based information by digitizing the information.
- E-commerce improves the brand image of the company.
- E-commerce helps organization to provide better customer services.
- E-Commerce helps to simplify the business processes and make them faster and efficient.
- E-Commerce reduces paper work a lot.

- E-Commerce increased the productivity of the organization. It supports "pull" type supply management. In "pull" type supply management, a business process starts when a request comes from a customer and it uses just-in-time manufacturing way.

Advantages to Customers

- 24x7 support. Customer can do transactions for the product or enquiry about any product/services provided by a company anytime, anywhere from any location. Here 24x7 refers to 24 hours of each seven days of a week.
- E-Commerce application provides user more options and quicker delivery of products.
- E-Commerce application provides user more options to compare and select the cheaper and better option.
- A customer can put review comments about a product and can see what others are buying or see the review comments of other customers before making a final buy.
- E-Commerce provides option of virtual auctions.
- Readily available information. A customer can see the relevant detailed information within seconds rather than waiting for days or weeks.
- E-Commerce increases competition among the organizations and as result organizations provides substantial discounts to customers.

Advantages to Society

- Customers need not to travel to shop a product thus less traffic on road and low air pollution.
- E-Commerce helps reducing cost of products so less affluent people can also afford the products.
- E-Commerce has enabled access to services and products to rural areas as well which are otherwise not available to them.
- E-Commerce helps government to deliver public services like health care, education, social services at reduced cost and in improved way.

Disadvantages

E-Commerce disadvantages can be broadly classified in two major categories:

1. Technical disadvantages
2. Non-Technical disadvantages

Technical disadvantages

- There can be lack of system security, reliability or standards owing to poor implementation of e-Commerce.
- Software development industry is still evolving and keeps changing rapidly.
- In many countries, network bandwidth might cause an issue as there is insufficient telecommunication bandwidth available.
- Special types of web server or other software might be required by the vendor setting the e-commerce environment apart from network servers.
- Sometimes, it becomes difficult to integrate E-Commerce software or website with the existing application or databases.
- There could be software/hardware compatibility issue as some E-Commerce software may be incompatible with some operating system or any other component.

Non-Technical disadvantages

- Initial cost: The cost of creating / building E-Commerce application in-house may be very high. There could be delay in launching the E-Commerce application due to mistakes, lack of experience.
- User resistance: User may not trust the site being unknown faceless seller. Such mistrust makes it difficult to make user switch from physical stores to online/virtual stores.
- Security/ Privacy: Difficult to ensure security or privacy on online transactions.
- Lack of touch or feel of products during online shopping.
- E-Commerce applications are still evolving and changing rapidly.
- Internet access is still not cheaper and is inconvenient to use for many potential customers like one living in remote villages.

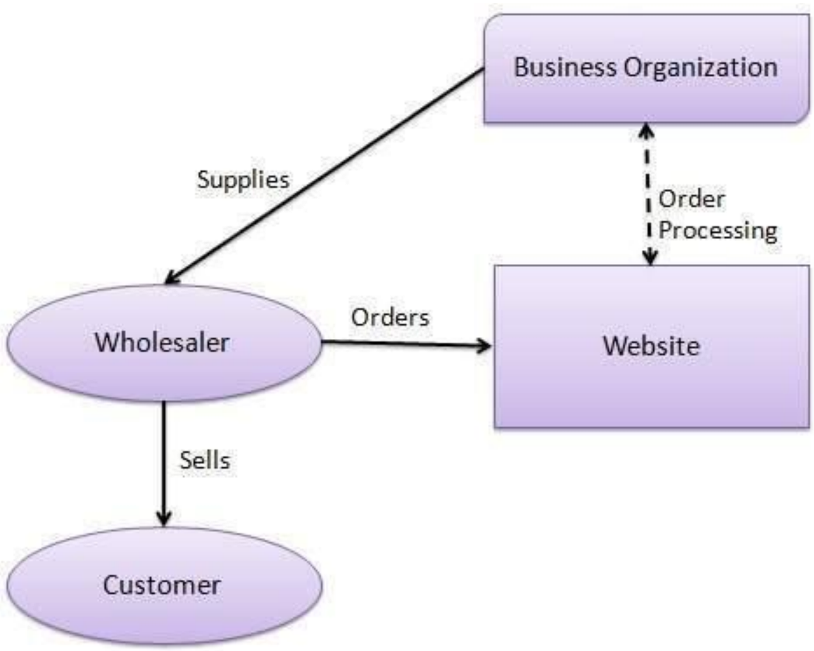
4. Explain briefly about E-Business model? (11 MARKS) (NOV 2016, 2017)

E-Commerce or Electronics Commerce business models can generally categorized in following categories.

1. Business - to - Business (B2B)
2. Business - to - Consumer (B2C)
3. Consumer - to - Consumer (C2C)
4. Consumer - to - Business (C2B)
5. Business - to - Government (B2G)
6. Government - to - Business (G2B)
7. Government - to - Citizen (G2C)

1. <u>Business - to - Business (B2B)</u>

Website following B2B business model sells its product to an intermediate buyer who then sells the product to the final customer. As an example, a wholesaler places an order from a company's website and after receiving the consignment, sells the end product to final customer who comes to buy the product at wholesaler's retail outlet.



B2B implies that seller as well as buyer is business entity. B2B covers large number of applications which enables business to form relationships with their distributors, resellers, suppliers etc.

Following are the leading items in B2B e-Commerce.

- Electronics
- Shipping and Warehousing
- Motor Vehicles
- Petrochemicals
- Paper
- Office products
- Food
- Agriculture

Key Technologies

Following are the key technologies used in B2B e-commerce −

- **Electronic Data Interchange (EDI)** − EDI is an inter organizational exchange of business documents in a structured and machine processable format.
- **Internet** − Internet represents World Wide Web or network of networks connecting computers across the world.
- **Intranet**− Intranet represents a dedicated network of computers within a single organization
- **Extranet** − Extranet represents a network where outside business partners, supplier or customers can have limited access to a portion of enterprise intranet/network.
- **Back-End Information System Integration** − Back End information systems are database management systems used to manage the business data.

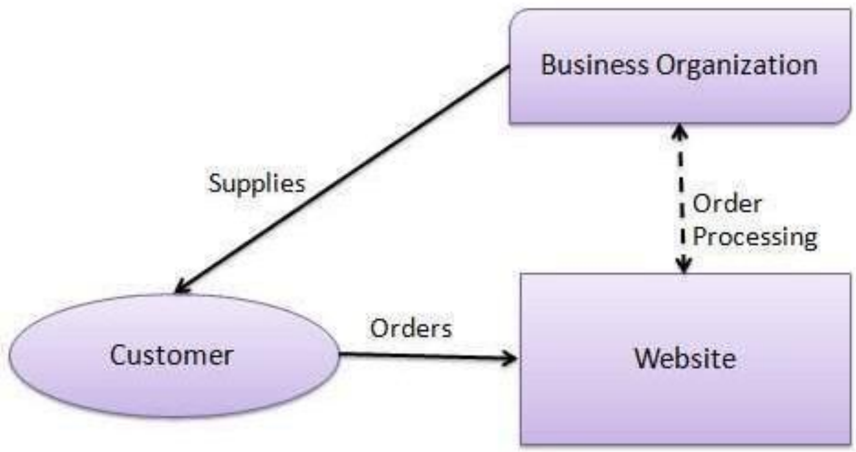
Architectural Models

Following are the architectural models in B2B e-commerce −

- **Supplier Oriented marketplace** − In this type of model, a common marketplace provided by supplier is used by both individual customers as well as business users. A supplier offers an e-stores for sales promotion.
- **Buyer Oriented marketplace** − In this type of model, buyer has his/her own market place or e-market. He invites suppliers to bid on product's catalog. A Buyer company opens a bidding site.
- **Intermediary Oriented marketplace** − In this type of model, an intermediary company runs a market place where business buyers and sellers can transact with each other.

2. Business - to - Consumer (B2C)

Website following B2C business model sells its product directly to a customer. A customer can view products shown on the website of business organization. The customer can choose a product and order the same. Website will send a notification to the business organization via email and organization will dispatch the product/goods to the customer.



In B2C Model, a consumer goes to the website, selects a catalog, orders the catalog and an email is sent to business organization. After receiving the order, goods would be dispatched to the customer. Following are the key features of a B2C Model

- Heavy advertising required to attract large no. of customers.
- High investment in terms of hardware/software.
- Support or good customer care service

Consumer Shopping Procedure

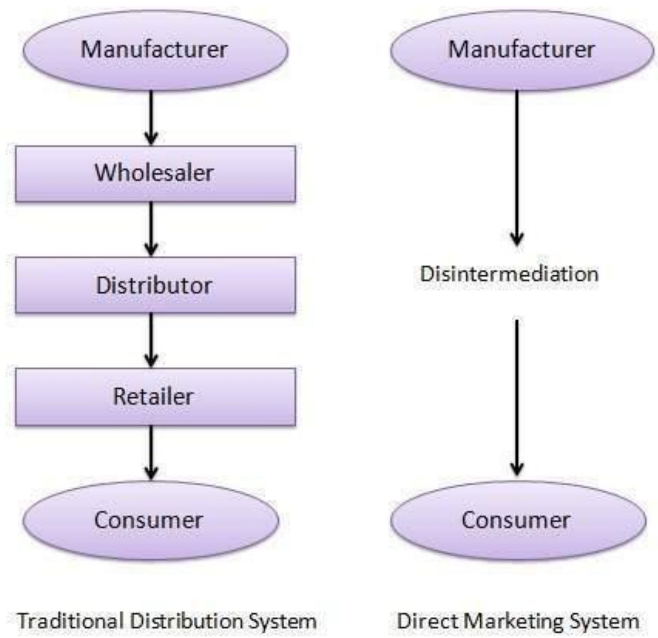
Following are the steps used in B2C e-commerce –

A consumer

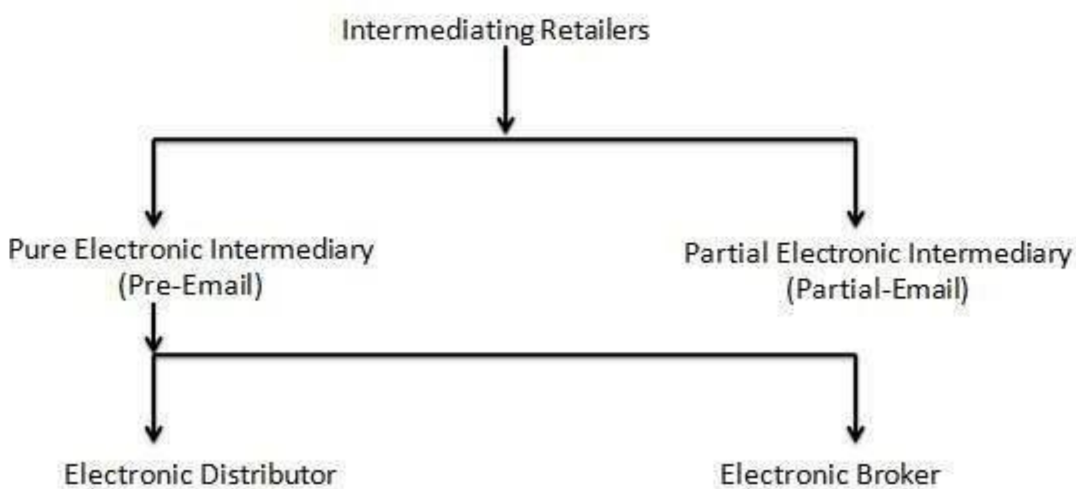
- determines the requirement.
- searches available items on the website meeting the requirment.
- compares similar items for price, delivery date or any other terms.
- gives the order.
- pays the bill.
- receives the delivered item and review/inspect them.
- consults the vendor to get after service support or returns the product if not satisfied with the delivered product.

Dis-intermediation and Re-intermediation

In traditional commerce, there are intermediating agents like wholesalers, distributors, retailers between manufacturer and consumer. In B2C website, manufacturer can sell products directly to consumers. This process of removal of business layers responsible for intermediary functions is called Disintermediation.

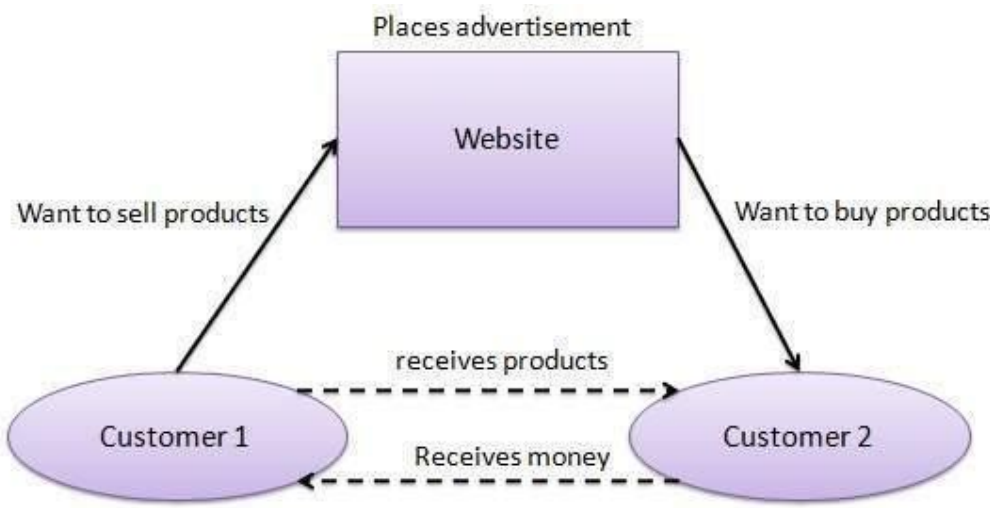


Now-a-days, a new electronic intermediary breed is emerging like e-mall and product selection agents are emerging. This process of shifting of business layers responsible for intermediary functions from traditional to electronic mediums is called Reintermediation.



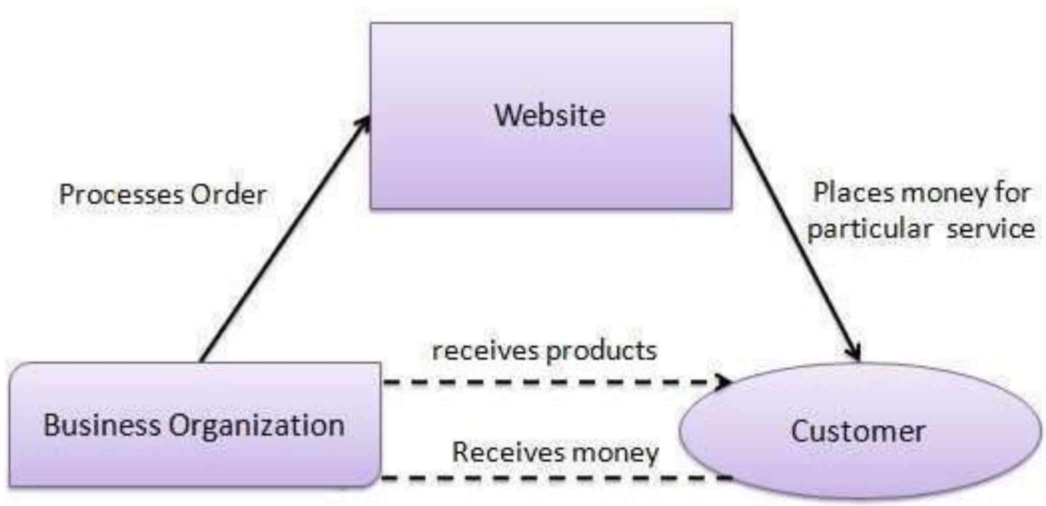
3. Consumer - to - Consumer (C2C)

Website following C2C business model helps consumer to sell their assets like residential property, cars, motorcycles etc. or rent a room by publishing their information on the website. Website may or may not charge the consumer for its services. Another consumer may opt to buy the product of the first customer by viewing the post/advertisement on the website.



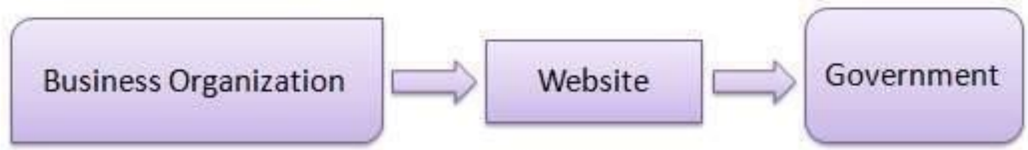
4. Consumer - to - Business (C2B)

In this model, a consumer approaches website showing multiple business organizations for a particular service. Consumer places an estimate of amount he/she wants to spend for a particular service. For example, comparison of interest rates of personal loan/ car loan provided by various banks via website. Business organization who fulfills the consumer's requirement within specified budget approaches the customer and provides its services.



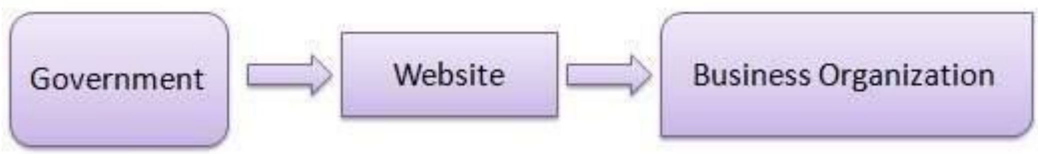
5. Business - to - Government (B2G)

B2G model is a variant of B2B model. Such websites are used by government to trade and exchange information with various business organizations. Such websites are accredited by the government and provide a medium to businesses to submit application forms to the government.



6. Government - to - Business (G2B)

Government uses B2G model website to approach business organizations. Such websites support auctions, tenders and application submission functionalities.



7. Government - to - Citizen (G2C)

Government uses G2C model website to approach citizen in general. Such websites support auctions of vehicles, machinery or any other material. Such website also provides services like registration for birth, marriage or death certificates. Main objectives of G2C website are to reduce average time for fulfilling people requests for various government services.

5. Discuss briefly about E-Marketing? (11 MARKS)

- E-Marketing is any marketing done online via websites or other online tools and resources.
- E-Marketing can include paid services while other methods are virtually free.
- A wide variety of e-Marketing methods are at your disposal, including: direct email, SMS/text messaging, blogs, webpages, banners, videos, images, ads, social media, search engines etc..
- Which e-marketing tools can assist?
  - Web, e-mail, databases, wireless and digital television.
- A type of direct marketing to a group of people using email with the purpose of acquiring new customers or convincing current customers to purchase something immediately.

Promoting a business by sending emails and newsletters is what we call email marketing. Today's marketers need to do more with less. They need to connect with their audience in a highly personalized way, while staying on budget. Marketers who are good at email marketing can connect with their customers in a highly targeted way. They will be successful in delivering ROI and revenue back to the business.

No marketing category has the longevity of email marketing. While some marketing trends come and go, email remains the most powerful channel available to the modern marketer.

Why should we do Email Marketing?

Email is a tool that nearly everyone uses today, and it continues to grow and be more prevalent in the lives of people around the world. There are three times more email accounts than there are Facebook & Twitter accounts combined. Many of the top marketers from some of the most successful companies across the world believe email is the #1 channel for growing your business.

As a marketer, you have many channels available to reach your audience, but with limited time and resources, you need to prioritize your efforts. Email Marketing is by far the most effective channel to attract, engage and connect with an audience to drive sales and revenue for your business.

E-Marketing Methods

1) Search engine marketing
2) Display Advertising
3) E-mail marketing
4) Interactive marketing
5) Blog marketing
6) Viral marketing

Email-Marketing- Newsletter

Newsletters are a shortened form of **newspaper** and **informational letter**. Generally used to describe a periodic publication distributed by e-mail to an **opt-in** list of subscribers. Newsletters are normally used by organizations or owners of a Website to communicate with their readers.

Some companies may sell targeted ads within their newsletters. Bear in mind, if you are going to send a Newsletter to subscribers then balance your **newsletter content to be 90% educational and 10% promotional**.


the treat

Wishing you a Happy & Healthy New Year!

Greetings,
I would like to start by saying Thank You. 2015 was an incredible year
We were able to serve thousands of victims of domestic violence and we wouldn't have been able to do it without you.

This year, we are celebrating 25 years of providing hope, inspiration, support and courage to those in our community affected by domestic abuse. The Retreat began in 1987 by a small group of community leaders who realized the need within the community. With initial funding from the East Hampton Rotary, the treat has been a safe haven for women and children and has been expanding its services ever since. 2012 will be a year-long celebration and we hope you will take part in the many different events and opportunities we'll be offering to support the agency so we can continue to be here for another 25 years.

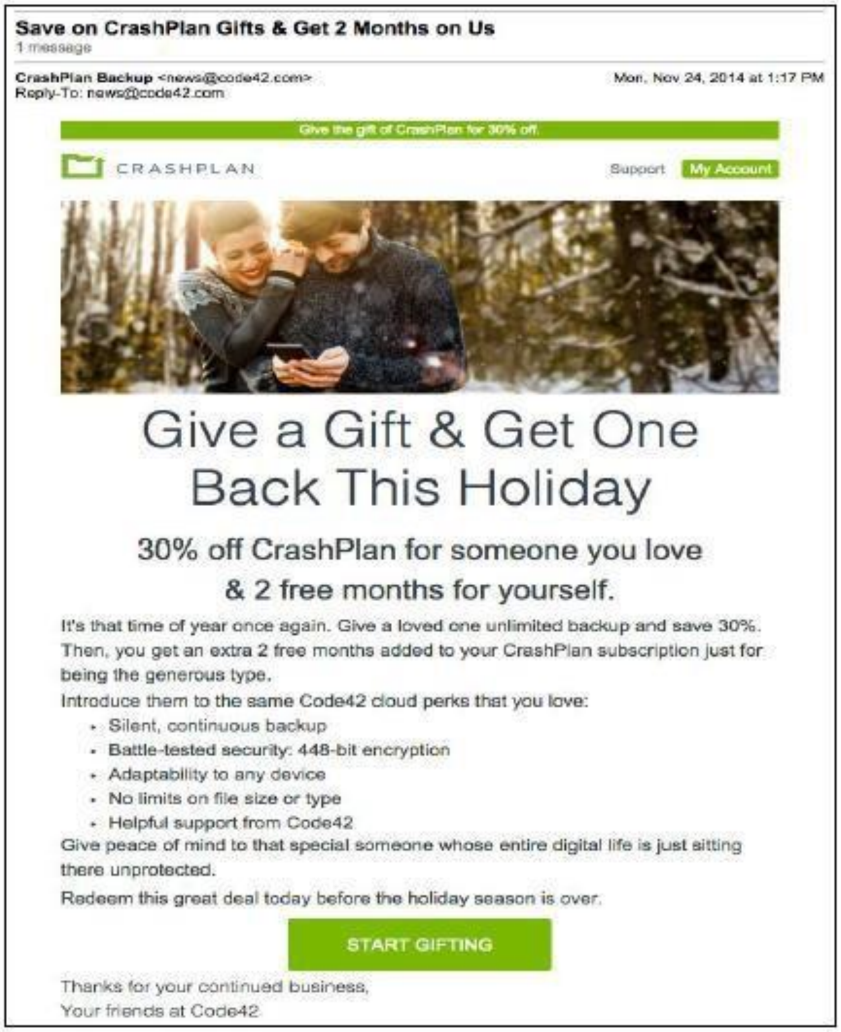
The women and children affected by domestic violence need you. The Retreat needs you. I look forward to working with you in the New Year.

Sincerely,

Nicole Ronar
Executive Director

Email-Marketing Promotional Newsletters

Be cautious with uncommon offer messages. This is the sort of stuff that is almost ensured to wind up in the "Promotional tab" in Gmail. When you send an exceptional offer, make sure you segment your lists first. You would prefer not to send a discount code to somebody who simply paid the maximum yesterday.



Email-Marketing-Event Invitation

Events and emails go hand in hand. Whether you host an open house, charity gala, webinar or a customer appreciation day, the best way to **promote your event** and invite guests is through email invitation. It is similar to what you have been doing for all your other emails. The following image shows a sample template of an email invitation.

Email Copywriting

Email marketing involves a unique form of copywriting that a lot of people, especially when just starting out, have some difficulty planning and executing. There are many similarities with other forms of copywriting, but there are also some unique opportunities and pitfalls as well. These articles will examine some of the elements of writing for email marketing and walk you through the basic steps of crafting your message.

Email-Marketing –Landing Pages

Email marketing is a powerful tool in itself, but it would be adding to the elegance with the use of landing pages. These pages are the ones which you put on your website that customers can link to from your email. Landing pages are an extensively detailed image of your email campaign with more info, more images, and even a purchase option, so recipients can buy what you're selling.

Email-Marketing-Automation

Scheduling

A better feature you would find in most of the service providers. It is very useful, if you are going to make a massive blast of discount on a Black Friday or on Happy New Year and so on. This feature will automatically send your email at the specified date and time.

Tracking

Tracking has made possible to track the result of your email campaign. It tells us the statistics of our email campaign. After sending emails to the list, tracking shows us the result of our campaign.

6. Explain briefly about the online payments and security? (11 MARKS)

Electronic - Payment Systems

E-Commerce or Electronics Commerce sites use electronic payment where electronic payment refers to paperless monetary transactions. Electronic payment has revolutionized the business processing by reducing paper work, transaction costs, labour cost. Being user friendly and less time consuming than manual processing, helps business organization to expand its market reach / expansion.

Some of the modes of electronic payments are following.

1. Credit Card
2. Debit Card
3. Smart Card
4. E-Money
5. Electronic Fund Transfer (EFT)

Credit Card

Payment using credit card is one of most common mode of electronic payment. Credit card is small plastic card with a unique number attached with an account. It has also a magnetic strip embedded in it which is used to read credit card via card readers. When a customer purchases a product via credit card, credit card issuer bank pays on behalf of the customer and customer has a certain time period after which he/she can pay the credit card bill. It is usually credit card monthly payment cycle. Following are the actors in the credit card system.

- The card holder - Customer
- The merchant - seller of product who can accept credit card payments.
- The card issuer bank - card holder's bank
- The acquirer bank - the merchant's bank
- The card brand - for example, visa or MasterCard.

Debit Card

Debit card, like credit card is a small plastic card with a unique number mapped with the bank account number. It is required to have a bank account before getting a debit card from the bank. The major difference between debit card and credit card is that in case of payment through debit card, amount gets deducted from card's bank account immediately and there should be sufficient balance in bank account for the transaction to get completed. Whereas in case of credit card there is no such compulsion.

Debit cards free customer to carry cash, cheques and even merchants accepts debit card more readily. Having restriction on amount being in bank account also helps customer to keep a check on his/her spendings.

Smart Card

Smart card is again similar to credit card and debit card in appearance but it has a small microprocessor chip embedded in it. It has the capacity to store customer work related/personal information. Smart card is also used to store money which is reduced as per usage.

Smart card can be accessed only using a PIN of customer. Smart cards are secure as they stores information in encrypted format and are less expensive/provides faster processing. Mondex and Visa Cash cards are examples of smart cards.

E-Money

E-Money transactions refers to situation where payment is done over the network and amount gets transferred from one financial body to another financial body without any involvement of a middleman. E-money transactions are faster, convenient and saves a lot of time.

Online payments done via credit card, debit card or smart card are examples of e-money transactions. Another popular example is e-cash. In case of e-cash, both customer and merchant both have to sign up with the bank or company issuing e-cash.

Electronic Fund Transfer

It is a very popular electronic payment method to transfer money from one bank account to another bank account. Accounts can be in same bank or different bank. Fund transfer can be done using ATM (Automated Teller Machine) or using computer.

Now a day, internet based EFT is getting popularity. In this case, customer uses website provided by the bank. Customer logins to the bank's website and registers another bank account. He/she then places a request to transfer certain amount to that account. Customer's bank transfers amount to other account if it is in same bank otherwise transfer request is forwarded to ACH (Automated Clearing House) to transfer amount to other account and amount is deducted from customer's account. Once amount is transferred to other account, customer is notified of the fund transfer by the bank.

E-Commerce - Security Systems

Security is an essential part of any transaction that takes place over the internet. Customer will lose his/her faith in e-business if its security is compromised.

Following are the essential requirements for safe e-payments/transactions −

- **Confidential** − Information should not be accessible to unauthorized person. It should not be intercepted during transmission.
- **Integrity** − Information should not be altered during its transmission over the network.
- **Availability** − Information should be available wherever and whenever requirement within time limit specified.
- **Authenticity** − There should be a mechanism to authenticate user before giving him/her access to required information.
- **Non-Repudiabiity** − It is protection against denial of order or denial of payment. Once a sender sends a message, the sender should not able to deny sending the message. Similary the receipient of message should not be able to deny receipt.
- **Encryption** − Information should be encrypted and decrypted only by authorized user.
- **Auditability** − Data should be recorded in such a way that it can be audited for integrity requirements.

Measures to ensure Security

Major security measures are following −

- **Encryption** − It is a very effective and practical way to safeguard the data being transmitted over the network. Sender of the information encrypt the data using a secret code and specified receiver only can decrypt the data using the same or different secret code.
- **Digital Signature** − Digital signature ensures the authenticity of the information. A digital signature is a e-signature authentic authenticated through encryption and password.

- **Security Certificates** – Security certificate is unique digital id used to verify identity of an individual website or user.

Security Protocols in Internet

Following are the popular protocols used over the internet which ensures security of transactions made over the internet.

Secure Socket Layer (SSL)

It is the most commonly used protocol and is widely used across the industry. It meets following security requirements –

- Authentication
- Encryption
- Integrity
- Non-reputability

"https://" is to be used for HTTP urls with SSL, where as "http:/" is to be used for HTTP urls without SSL.

Secure Hypertext Transfer Protocol (SHTTP)

SHTTP extends the HTTP internet protocol with public key encryption, authentication and digital signature over the internet. Secure HTTP supports multiple security mechanism providing security to end users. SHTTP works by negotiating encryption scheme types used between client and server.

Secure Electronic Transaction

It is a secure protocol developed by MasterCard and Visa in collaboration. Thereoritically, it is the best security protocol. It has following components –

- **Card Holder's Digital Wallet Software** – Digital Wallet allows card holder to make secure purchases online via point and click interface.
- **Merchant Software** – This software helps merchants to communicate with potential customers and financial institutions in secure manner.
- **Payment Gateway Server Software** – Payment gateway provides automatic and standard payment process. It supports the process for merchant's certificate request.
- **Certificate Authority Software** – This software is used by financial institutions to issue digital certificates to card holders and merchants and to enable them to register their account agreements for secure electronic commerce.

7. Explain in detail about the N-tier Architecture and its components? (11 MARKS) (APR 2016)

In software engineering, multitier architecture (often referred to as n-tier architecture) or multilayered architecture is a client–server architecture in which presentation, application processing, and data management functions are physically separated. The most widespread use of multitier architecture is the three-tier architecture.

N-tier application architecture provides a model by which developers can create flexible and reusable applications. By segregating an application into tiers, developers acquire the option of modifying or adding a specific layer, instead of reworking the entire application.

A three-tier architecture is typically composed of a

1. Presentation tier,
2. Business/ logic tier, and
3. Data storage tier.

N-tier architectures try to separate the components into different tiers/layers

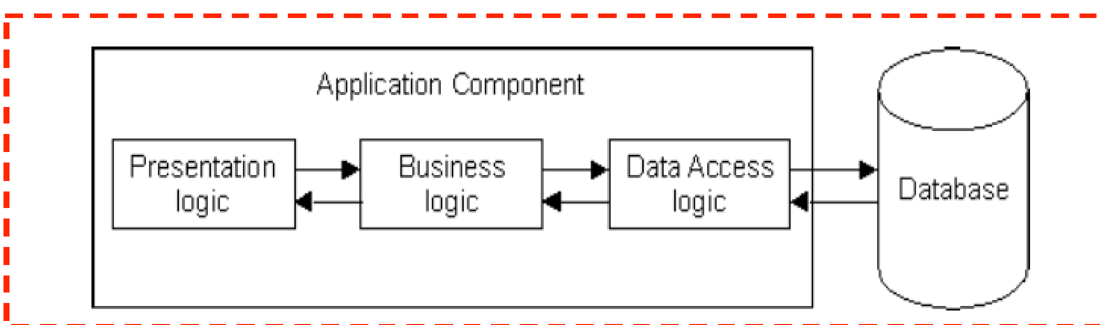
- Tier: physical separation
- Layer: logical separation

Common layers

In a logical multilayered architecture for an information system with an object-oriented design, the following four are the most common:

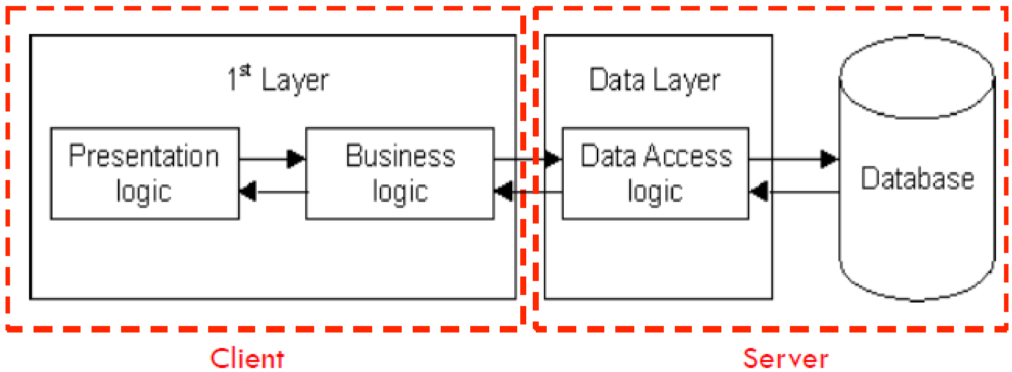
1. **Presentation layer** (UI layer, view layer, presentation tier in multitier architecture)
2. **Application layer** (service layer **or** GRASP **Controller** Layer)
3. **Business layer** (business logic layer **(BLL),** domain layer)
4. **Data access layer** (persistence layer, **logging,** networking, and other services which are required to support a particular business layer)

1-Tier Architecture



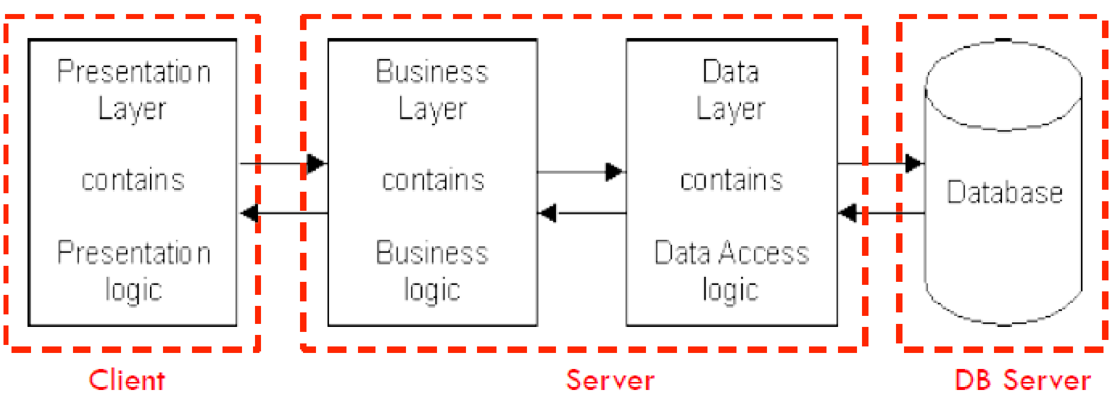
- All 3 layers are on the same machine
  - All code and processing kept on a single machine
- Presentation, Logic, Data layers are tightly connected
  - Scalability: Single processor means hard to increase volume of processing
  - Portability: Moving to a new machine may mean rewriting everything
  - Maintenance: Changing one layer requires changing other layers

2-Tier Architecture



- Database runs on Server
  - Separated from client
  - Easy to switch to a different database
- Presentation and logic layers still tightly connected
  - Heavy load on server
  - Potential congestion on network
  - Presentation still tied to business logic

3-Tier Architecture

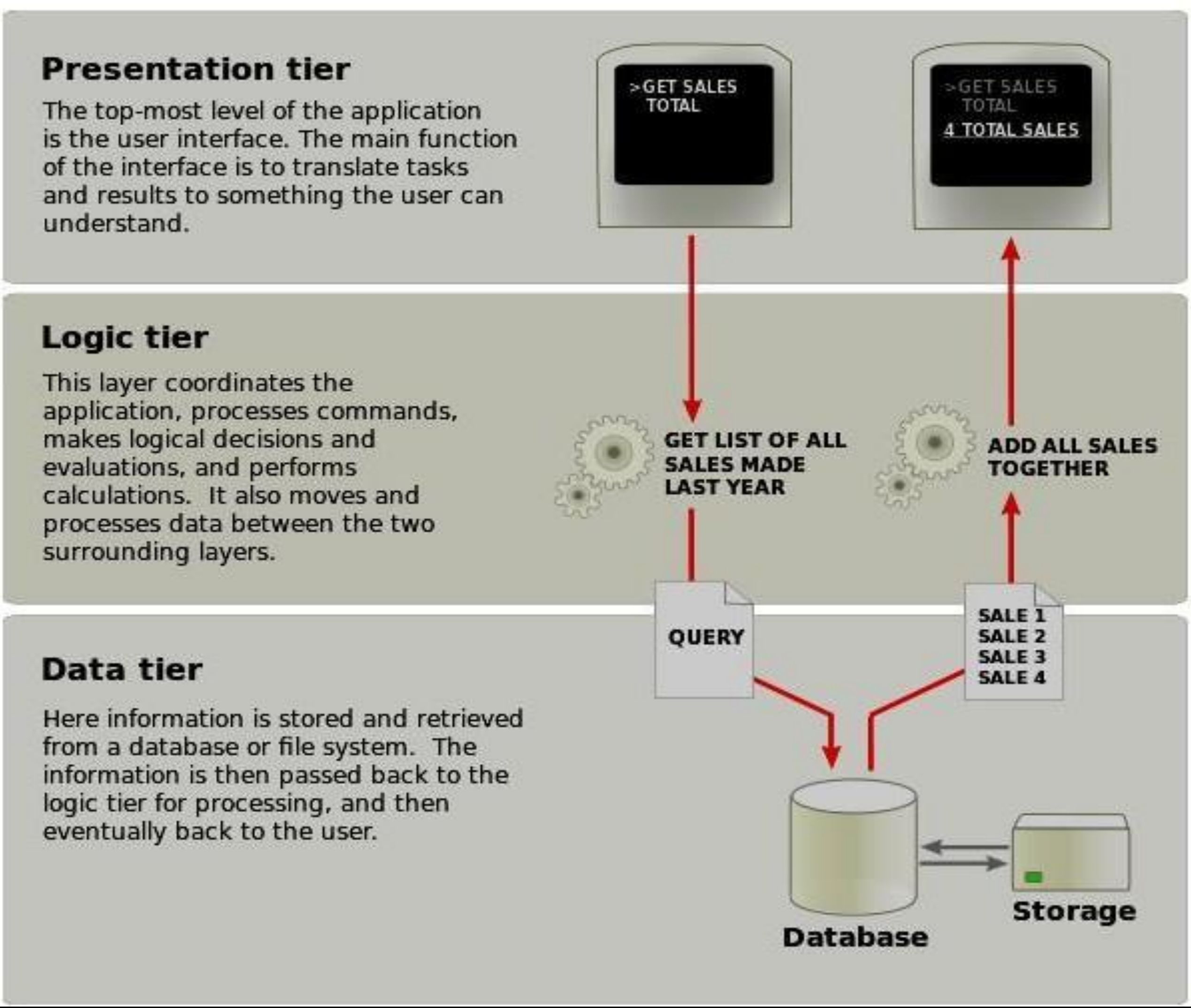


- Each layer can potentially run on a different machine
  - Presentation, logic, data layers disconnected

Three-tier architecture

- Three-tier architecture is a client–server software architecture pattern in which the user interface (presentation), functional process logic ("business rules"), computer data storage and data access are developed and maintained as independent modules, most often on separate platforms.
- Apart from the usual advantages of modular software with well-defined interfaces, the three-tier architecture is intended to allow any of the three tiers to be upgraded or replaced independently in response to changes in requirements or technology. For example, a change of operating system in the presentation tier would only affect the user interface code.
- Typically, the user interface runs on a desktop PC or workstation and uses a standard graphical user interface, functional process logic that may consist of one or more separate modules running on a workstation or application server, and an RDBMS on a database server or mainframe that contains the computer data storage logic. The middle tier may be multi-tiered itself (in which case the overall architecture is called an "n-tier architecture").

Three-tier architecture:



Presentation tier

This is the topmost level of the application. The presentation tier displays information related to such services as browsing merchandise, purchasing and shopping cart contents. It communicates with other tiers by which it puts out the results to the browser/client tier and all other tiers in the network. In simple terms, it is a layer which users can access directly (such as a web page, or an operating system's GUI).

Application tier (business logic, logic tier, or middle tier)

The logical tier is pulled out from the presentation tier and, as its own layer, it controls an application's functionality by performing detailed processing.

Data tier

The data tier includes the data persistence mechanisms (database servers, file shares, etc.) and the data access layer that encapsulates the persistence mechanisms and exposes the data. The data access layer should provide an API to the application tier that exposes methods of managing the stored data without exposing or creating dependencies on the data storage mechanisms. Avoiding dependencies on the storage mechanisms allows for updates or changes without the application tier clients being affected by or even aware of the change. As with the separation of any tier, there are costs for implementation and often costs to performance in exchange for improved scalability and maintainability.

The 3-Tier Architecture for Web Apps

Presentation Layer

Static or dynamically generated content rendered by the browser (front-end)

Logic Layer

A dynamic content processing and generation level application server, e.g., Java EE, ASP.NET, PHP, ColdFusion platform (middleware)

Data Layer

A database, comprising both data sets and the database management system or RDBMS software that manages and provides access to the data (back-end)

3-Tier Architecture – Advantages

Independence of Layers

- Easier to maintain
- Components are reusable
- Faster development (division of work)
- Web designer does presentation
- Software engineer does logic
- DB admin does data model

8. Discuss about design and working principles of search engine? (11 MARKS) (NOV 2016, 2017)

Search Engine refers to a huge database of internet resources such as web pages, newsgroups, programs, images etc. It helps to locate information on World Wide Web. User can search for any information by passing query in form of keywords or phrase. It then searches for relevant information in its database and return to the user.

Search Engine Components

Generally there are three basic components of a search engine as listed below:

1. Web Crawler
2. Database
3. Search Interfaces

Web crawler

It is also known as spider or bots. It is a software component that traverses the web to gather information.

Database

All the information on the web is stored in database. It consists of huge web resources.

Search Interfaces

This component is an interface between user and the database. It helps the user to search through the database.

Search Engine Working

Web crawler, database and the search interface are the major component of a search engine that actually makes search engine to work. Search engines make use of Boolean expression AND, OR, NOT to restrict and widen the results of a search. Following are the steps that are performed by the search engine:

- The search engine looks for the keyword in the index for predefined database instead of going directly to the web to search for the keyword.
- It then uses software to search for the information in the database. This software component is known as web crawler.
- Once web crawler finds the pages, the search engine then shows the relevant web pages as a result. These retrieved web pages generally include title of page, size of text portion, first several sentences etc.

These search criteria may vary from one search engine to the other. The retrieved information is ranked according to various factors such as frequency of keywords, relevancy of information, links etc. User can click on any of the search results to open it.

How Search Engine Works?

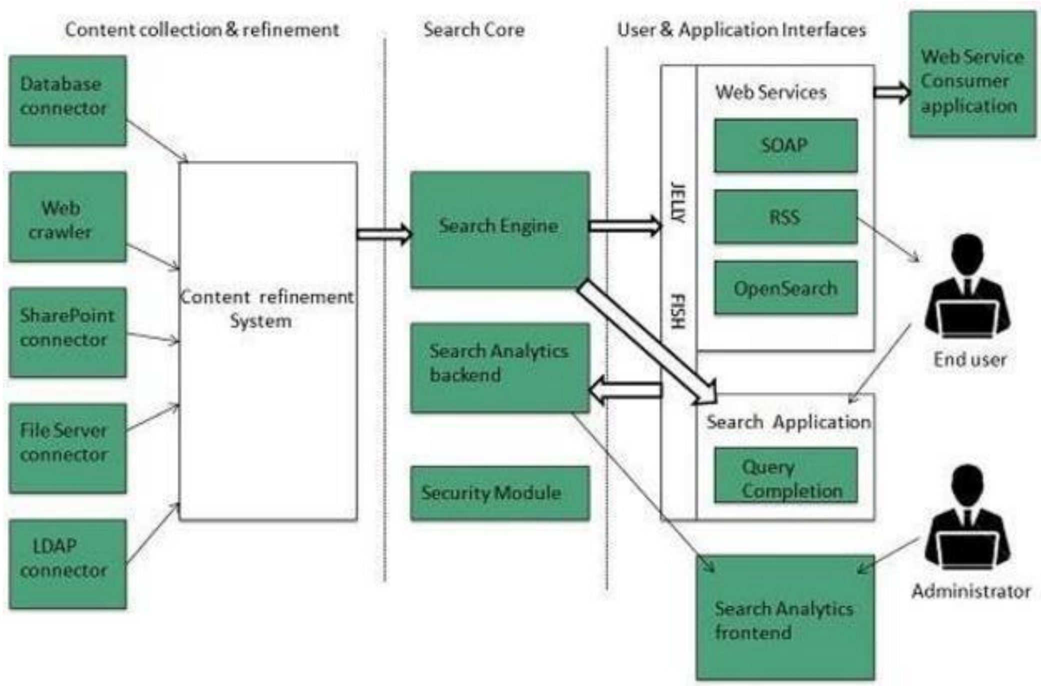
Search engines perform several activities in order to deliver search results.

- **Crawling -** Process of fetching all the web pages linked to a website. This task is performed by a software, called a **crawler** or a **spider** (or Googlebot, in case of Google).
- **Indexing -** Process of creating index for all the fetched web pages and keeping them into a huge database from where it can later be retrieved. Essentially, the process of indexing is identifying the words and expressions that best describe the page and assigning the page to particular keywords.
- **Processing -** When a search request comes, the search engine processes it, i.e. it compares the search string in the search request with the indexed pages in the database.
- **Calculating Relevancy -** It is likely that more than one page contains the search string, so the search engine starts calculating the relevancy of each of the pages in its index to the search string.
- **Retrieving Results -** The last step in search engine activities is retrieving the best matched results. Basically, it is nothing more than simply displaying them in the browser.

Architecture

The search engine architecture comprises of the three basic layers listed below:

- Content collection and refinement.
- Search core
- User and application interfaces



Search Engine Processing

Indexing Process

Indexing process comprises of the following three tasks:

- Text acquisition
- Text transformation
- Index creation

Text Acquisition

It identifies and stores documents for indexing.

Text Transformation

It transforms document into index terms or features.

Index Creation

It takes index terms created by text transformations and create data structures to support fast searching.

Query Process

Query process comprises of the following three tasks:

- User interaction
- Ranking
- Evaluation

User Interaction

It supports creation and refinement of user query and displays the results.

Ranking

It uses query and indexes to create ranked list of documents.

Evaluation

It monitors and measures the effectiveness and efficiency. It is done offline.

Examples

Following are the several search engines available today:

Search Engine	**Description**
Google	It was originally called **BackRub.** It is the most popular search engine globally.
Bing	It was launched in 2009 by **Microsoft.** It is the latest web-based search engine that also delivers Yahoo's results.
Ask	It was launched in 1996 and was originally known as **Ask Jeeves.** It includes support for match, dictionary, and conversation question.
AltaVista	It was launched by **Digital Equipment Corporation** in 1995. Since 2003, it is powered by Yahoo technology.
AOLSearch	It is powered by Google.
LYCOS	It is top 5 internet portal and 13th largest online property according to Media Matrix.
Alexa	It is subsidiary of Amazon and used for providing website traffic information.

9. Explain in detail about Search Engine Optimization (SEO)? (6 MARKS)

Search Engine Optimization (SEO) is the activity of optimizing web pages or whole sites in order to make them search engine friendly, thus getting higher positions in search results.

<u>What is SEO Copywriting?</u>

SEO Copywriting is the technique of writing viewable text on a web page in such a way that it reads well for the surfer, and also targets specific search terms. Its purpose is to rank highly in the search engines for the targeted search terms.

Along with viewable text, SEO copywriting usually optimizes other on-page elements for the targeted search terms. These include the Title, Description, Keywords tags, headings, and alternative text.

The idea behind SEO copywriting is that search engines want genuine content pages and not additional pages often called "doorway pages" that are created for the sole purpose of achieving high rankings.

<u>What is Search Engine Rank?</u>

When you search any keyword using a search engine, it displays thousands of results found in its database. A page ranking is measured by the position of web pages displayed in the search engine results. If a search engine is putting your web page on the first position, then your web page rank will be number 1 and it will be assumed as the page with the highest rank.

SEO is the process of designing and developing a website to attain a high rank in search engine results.

<u>What is On-Page and Off-page SEO?</u>

Conceptually, there are two ways of optimization:

- **On-Page SEO -** It includes providing good content, good keywords selection, putting keywords on correct places, giving appropriate title to every page, etc.
- **Off-Page SEO -** It includes link building, increasing link popularity by submitting open directories, search engines, link exchange, etc.

SEO techniques are classified into two broad categories:

- **White Hat SEO -** Techniques that search engines recommend as part of a good design.
- **Black Hat SEO -** Techniques that search engines do not approve and attempt to minimize the effect of. These techniques are also known as spamdexing.

<u>SEO - Optimized Keywords</u>

A keyword is a term that is used to match with the query a person enters into a search engine to find specific information. Most people enter search phrases that consist of two to five words. Such phrases may be called search phrases, keyword phrases, query phrases, or just keywords. Good keyword phrases are specific and descriptive.

The following concepts related to keywords, help in optimizing the keywords on a web page.

- Keyword Frequency
- Keyword Weight
- Keyword Proximity
- Keyword Prominence
- Keyword Placement

A list of places where you should try to use your main keywords.

- ✓ Keywords in the <title> tag(s).
- ✓ Keywords in the <meta name="description">.
- ✓ Keywords in the <meta name="keyword">.
- ✓ Keywords in <h1> or other headline tags.
- ✓ Keywords in the keywords link tags.
- ✓ Keywords in the body copy.
- ✓ Keywords in alt tags.
- ✓ Keywords in <!-- insert comments here> comments tags.
- ✓ Keywords in the URL or website address.

SEO - Optimized Metatags

There are two important meta tags:

- Meta description tags
- Meta keyword tags

SEO - Title Optimization

An HTML TITLE tag is put inside the head tag. The page title (not to be confused with the heading for a page) is what is displayed in the title bar of your browser window, and is also what is displayed when you bookmark a page or add it to your browser Favorites.

SEO - Optimized Anchor

Use descriptive anchor text for all your text links. Most search engines consider anchor text of incoming links when ranking pages.

SEO Content Writing (Copy Writing)

SEO Content Writing (also referred as SEO Copy writing), involves the process of integrating keywords and informative phrases which make up the actual content of your website.

10. Explain multimedia Authoring and User Interface Design? **(11 MARKS) (APR 2017)**

Multimedia authoring systems are designed with two primary target users:

They are

i. Professionals who prepare documents, audio or sound tracks, and full motion video clips for wide distribution.

ii. Average business users preparing documents, audio recordings, or full motion video clips for stored messages' or presentations.

The authoring system covers user interface. The authoring system spans issues such as data access, storage structures for individual components embedded in a document, the user's ability to browse through stored objects, and so on.

Most authoring systems are managed by a control application.

Design Issues for Multimedia Authoring

Enterprise wide standards should be set up to ensure that the user requirements are fulfilled with good quality and made the objects transferable from one system to another. So standards must be set for a number of design issues

1. Display resolution
2. Data formula for capturing data
3. Compression algorithms
4. Network interfaces
5. Storage formats.

Display resolution

A number of design issues must be considered for handling different display outputs. They are:

(a) Level of standardization on display resolutions.

(b) Display protocol standardization.

(c) Corporate norms for service degradations

(d) Corporate norms for network traffic degradations as they relate to resolution issues Setting norms will be easy if the number of different work station types, window managers, and monitor resolutions are limited in number. But if they are more in number, setting norms will be difficult. Another consideration is selecting protocols to use. Because a number of protocols have emerged, including AVI, video, Quick Time and so on. So, there should be some level of convergence that allows these three display protocols to exchange data and allow viewing files in other formats.

File Format and Data Compression Issues

There are variety of data formats available for image, audio, and full motion video objects.

Since the varieties are so large, controlling them becomes difficult. So we should not standardize on a single format. Instead, we should select a set for which reliable conversion application tools are available.

Another key design Issue is to standardize on one or two compression formula for each type of data object. For example for facsimile machines, CCITT Group 3 and 4 should be included in the selected standard. Similarly, for full motion video, the selected standard should include MPEG and its derivatives such as MPEG.

While doing storage, it is useful to have some information (attribute information) about the object itself available outside the object to allow a user to decide if they need to access the object data. one of such attribute information are:

(i) Compression type

(ii) Size of the object

(iii) Object orientation

(iv) Data and time of creation

(v) Source file name

(vi) Version number

(vii) Required software application to display or playback the object.

Service degradation policies

Setting up Corporate norms for network traffic degradation is difficult as they relate to resolution Issues:

To address these design issues, several policies are possible. They are:

1. Decline further requests with a message to try later.
2. Provide the playback server but at a lower resolution.
3. Provide the playback service at full resolution but, in the case of sound and full motion video, drop intermediate frames.

Design Approach to Authoring

Designing an authoring system spans a number of design issues. They include:

Hypermedia application design specifics, User Interface aspects, Embedding/Linking streams of objects to a main document or presentation, Storage of and access to multimedia objects. Playing back combined streams in a synchronized manner.

A good user interface design is more important to the success of hypermedia applications.

Types of Multimedia Authoring Systems

There are varying degrees of complexity among the authoring systems. For example, dedicated authoring systems that handle only one kind of an object for a single user is simple, where as programmable systems are most complex.

Dedicated Authority Systems

Dedicated authoring systems are designed for a single user and generally for single streams. Designing this type of authoring system is simple, but if it should be capable of combining even two object streams, it becomes complex. The authoring is performed on objects captured by the local video camera and image scanner or an objects stored in some form of multimedia object library. In the case of dedicated authoring system, users need not to be experts in multimedia or a professional artist. But the dedicated systems should be designed in such a way that. It has to provide user interfaces that are extremely intuitive and follow real-world metaphors.

A structured design approach will be useful in isolating the visual and procedural design components.

TimeLine –based authoring

In a timeline based authoring system, objects are placed along a timeline. The timeline can be drawn on the screen in a window in a graphic manner, or it created using a script in a mann.er similar to a project plan. But, the user must specify a resource object and position it in the timeline.

In most timeline based approaches, once the multimedia object has been captured in a timeline. It is fixed in location and cannot be manipulated easily. So, a single timeline causes loss of information about the relative time lines for each individual object.

Structured Multimedia Authoring

A structured multimedia authoring approach was presented by Hardman. It is an evolutionary approach based on structured object-level construction of complex presentations. This approach consists of two stages:

(i) The construction of the structure of a presentation.

(ii) Assignment of detailed timing constraints.

A successful structured authoring system must provide the following capabilities for navigating through the structure of presentation.

1. Ability to view the complete structure.
2. Maintain a hierarchy of objects.
3. Capability to zoom down to any specific component.
4. View specific components in part or from start to finish.

5. Provide a running status of percentage full of the designated length of the presentation.
6. Clearly show the timing relations between the various components.
7. Ability to address all multimedia types including text, image, audio, video and frame based digital images.

Programmable Authoring Systems

Early structured authoring tools were not able to allow the authors to express automatic function for handling certain routine tasks. But, programmable authoring system based improved in providing powerful functions based on image processing and analysis and embedding program interpreters to use image-processing functions.

The capability of this authoring system is enhanced by Building user programmability in the authoring tool to perform the analysis and to manipulate the stream based on the analysis results and also manipulate the stream based on the analysis results. The programmability allows the following tasks through the program interpreter rather than manually. Return the time stamp of the next frame. Delete a specified movie segment. Copy or cut a specified movie segment to the clip board. Replace the current segment with clip board contents.

Multisource Multi-user Authoring Systems

We can have an object hierarchy in a geographic plane; that is, some objects may be linked to other objects by position, while others may be independent and fixed in position".

We need object data, and information on composing it. Composing means locating it in reference to other objects in time as Well as space.

Once the object is rendered (display of multimedia object on the screen) the author can manipulate it and change its rendering information must be available at the same time for display. If there are no limits on network bandwidth and server performance, it would be possible to assemble required components on cue at the right time to be rendered.

In addition to the multi-user compositing function. A multi user authoring system must provide resource allocation and scheduling of multimedia objects.

Telephone authoring systems

There is an application where the phone is linking into multimedia electronic mail application

1. Telephone can be used as a reading device by providing fill text to-speech synthesis capability so that a user on the road can have electronic mail messages read out on the telephone.
2. The phone can be used for voice command input for setting up and managing voice mail messages. Digitized voice clips are captured via the phone and embedded in electronic mail messages.

3. As the capability to recognize continuous speech is deploy phones can be used to create electronic mail messages where the voice is converted to ASCII text on the fly by high-performance voice recognition engines.

Phones provide a means of using voice where the alternative of text on a screen is not available. A phone can be used to provide interactive access to electronic mail, calendar information databases, public information database and news reports, electronic newspapers and a variety of other applications. Integrating of all these applications in a common authoring tool requires great skill in planning.

The telephone authoring systems support different kinds of applications.
Some of them are:

1. Workstation controls for phone mail.
2. Voice command controls for phone mail.
3. Embedding of phone mail in electric mail.

Common UI and Application Integration

Microsoft Windows has standardized the user interface for a large number of applications by providing standardization at the following levels: Overall visual look and feel of the application windows

This standardization level makes it easier for the user to interact with applications designed for the Microsoft Windows operational environment. Standardization is being provided for Object Linking and Embedding (OLE), Dynamic Data Exchange (DOE), and the Remote Procedure Call (RPC).

Data Exchange

The Microsoft Windows Clipboard allows exchanging data in any format. It can be used to exchange multimedia objects also. We can cut and copy a multimedia objects in one document and pasting in another. These documents can be opened under different applications. The windows clipboard allows the following formats to be stored:

- Text Bitmap
- Image Sound
- Video (AVI format)

User Interface Design

Multimedia applications contain user interface design. There are four kinds of user interface development tools. They are

1. Media editors
2. An authoring application
3. Hypermedia object creation
4. Multimedia object locator and browser

A media editor is an application responsible of the creation and editing of a specific multimedia object such as an image, voice, or Video object. Any application that allows the user to edit a multimedia object contains a media editor. Whether the object is text, voice, or full-motion video, the basic functions provided by the editor are the same: create, delete, cut, copy, paste, move, and merge.

Navigation through the application

Navigation refers to the sequence in which the application progresses and objects are created, searched and used.

Navigation can be of three modes:

1. Direct: It is completely predefined. In this case, the user needs to know what to expect with successive navigation actions.
2. Free-form mode: In this mode~ the user determines the next sequence of actions.
3. Browse mode: In this mode, the user does not know the precise question and wants to get general information about a particular topic. It is a very common mode in application based on large volumes of non-symbolic data. This mode allows a user to explore the databases to support the hypothesis.

Designing user Interfaces

User Interface should be designed by structured following design guidelines as follows:

1. Planning the overall structure of the application
2. Planning the content of the application
3. Planning the interactive behavior
4. Planning the look and feel of the application